Politecnico di Milano

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Group No.
5

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1. Introduction

This report summarises the work carried out during the Aerospace Technologies Laboratory. The aim of the laboratory was to analyse and investigate the advantages and the issues of 3D printing using a lunar regolith simulant. Using *Autodesk Fusion®*, a bar with XY orientation and a cylinder with a gyroid lattice structure were designed.
The modelling parameters are reported below.

Flexural bars	X	Y	Z
Size	40 mm	5 mm	3 mm

Cylinder			*Offset (2 base-faces)*			*Gyroid infill*
Dimension	*Diameter*	*Height*	*Thickness*	*Blend distance*	*Solidity*	*Uniform*
Size	8 mm	16 mm	2 mm	1 mm	1.00	2 mm - 60 %

Using the slicing software *Chitubox®*, the process parameters for printing were set. With a *layer height* of 30 µm, four XY bars were printed, three of which were sintered, together with three cylinders, two of which were sintered.

To best address the several sections of this report, the group split into sub-teams, each of which covered a different topic in greater detail: materials, process parameters, quality of the manufactured components and mechanical properties.
Each topic naturally required different investigation techniques: for the materials, the supplied data were compared and examined in depth with respect to the specific consequences of the regolith composition. For the process parameters, the ones we considered most critical were highlighted, following the R.A.M.S. analysis. Quality was assessed through the analysis of microscope images, using several software tools, including MATLAB® and ImageJ®. Finally, the mechanical properties were tested in the laboratory with dedicated equipment: the bars were tested in flexure and the cylinders in compression.
Once the report was completed, all sub-teams worked together on writing the slides for the oral presentation.

2. R.A.M.S. Analysis of the Manufacturing Process

2.1. Materials

The feedstock used for 3D printing is a custom slurry made of 30% custom resin, obtained from oligomers and photopolymers, and 70% *LHS-1*, a solid powder blend designed to simulate the behaviour of lunar regolith.[1]
To control the manufacturing process and better understand the characteristics and properties of the final product, it is useful to characterise both components. The composition of the *LHS-1* solid blend is derived from samples collected during lunar exploration missions. It consists mainly of anorthosite particles (74.4%) and glass-rich basalt (24.7%). Chemically, it is made up of several oxides, including SiO_2, Al_2O_3 and CaO. The composition has some direct effects on the manufacturing process: the oxides absorb and scatter UV light and interfere with the photopolymerisation of the resin. Exposure time therefore becomes a critical process parameter.
Besides the chemical composition, particle size is also a key parameter in determining the physical (thermal and mechanical) properties of the slurry, which in turn affect the 3D printing results. For this reason, the measured particle size distribution was compared with the one provided by the manufacturer, revealing a discrepancy. In fact, all particles in the sample are smaller than 8.710 µm, whereas an expected fraction below 12% was reported. A larger amount of fine particles leads to a higher total specific surface area, a higher viscosity and a lower polymerisation rate. As a consequence, more resin is required and the likelihood of agglomeration, and hence of printing defects, increases, as will also be shown in the following sections.

2.2. Process Parameters

The process parameters set for printing are part of the project R.A.M.S. analysis[2]; the parameters identified as critical are reported below.

- Reliability (**R**.A.M.S.)
  The printing process must operate while minimising defects in the manufactured parts over a defined period of time. Processing times are therefore an essential element of reliability: exposure time is adopted as the critical process parameter (set to 15 s). Another key aspect is reducing the likelihood of printer failure. By selecting moderate *lift* and *retract* speeds (40 and 80 mm/min, respectively), wear of the printer mechanical arms during the production cycle is reduced, lowering the probability of machine failure.

- Availability (R.**A**.M.S.)
  The 3D printer must be operational whenever requested by the operator. Excessive *lift* and *retract* distances increase printing time and therefore reduce machine availability; the *lifting distance* is thus chosen as the critical parameter (set to 10 mm). Similarly, the total rest time between one layer and the next, given by the combination of *rest time before lift*, *after lift* and *after retract* (21 s of waiting time in total), increases potential machine downtime.

- Maintainability (R.A.**M**.S.)
  Simple and targeted interventions on the machine and the process must be possible. Appropriate *lift* and *retract* speeds, such as those mentioned above, help keep the z-axis correctly calibrated during printing, reducing the need for realignment interventions.

[1] Percentages are by weight.
[2] Printer used: *Elegoo Mars 4 DLP*[*]

- Safety (R.A.M.**S**.)
  Minimisation of risk for users, operators, equipment and/or components. An incorrect power level either harms the operator, if too high, or leads to incomplete curing of the resin, if too low: hence the light source power (18 W according to the *Elegoo Mars 4 DLP®* datasheet) is taken as the critical parameter for safety in the work area.

It is worth noting that each parameter may affect a different part of the R.A.M.S. analysis depending on how it is interpreted. Consider, for example, the *transition layer time*: it improves reliability by reducing possible mechanical defects between the base and the body of the part, but at the same time it increases the exposure time, negatively affecting availability.

3. Quality of the Manufactured Components

To assess the quality of the manufactured components, their dimensional features and defects were analysed through microscope image analysis. Mechanical strength tests were also performed. The assessment also took into account the critical process parameters identified above.

3.1. Dimensional Tolerances

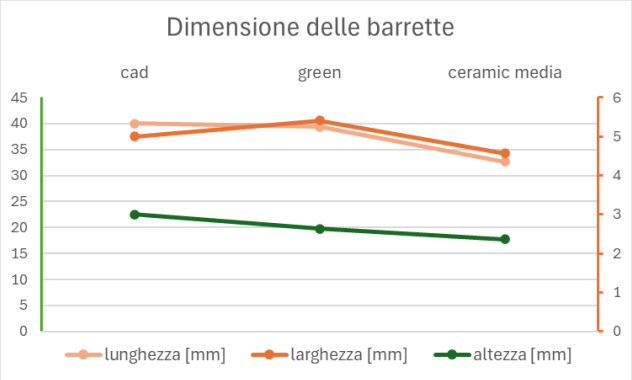
For the component analysis, data were acquired with optical and scanning electron microscopes[3], allowing observations at different scales. Data processing was performed with MATLAB® and ImageJ®, used individually or together to exploit their specific features and obtain the required data with the highest possible accuracy.

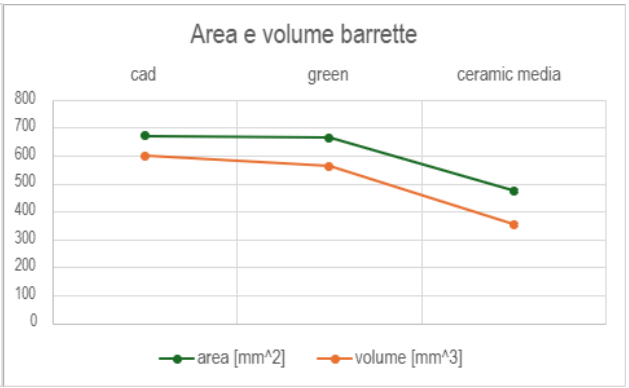
The charts show the nominal dimensions of the bar set in the CAD model and the experimental measurements taken with a digital caliper. The printed dimensions do not match those of the green (unsintered) specimen: length and height are smaller than nominal, whereas width is larger. Computing the percentage shrinkage between model and specimen, a negative shrinkage (i.e. an expansion) is observed in width. Possible causes of the CAD-to-green difference lie in the process parameters discussed in the R.A.M.S. analysis, in particular the rest time between successive layers and the correct alignment of the z-axis.

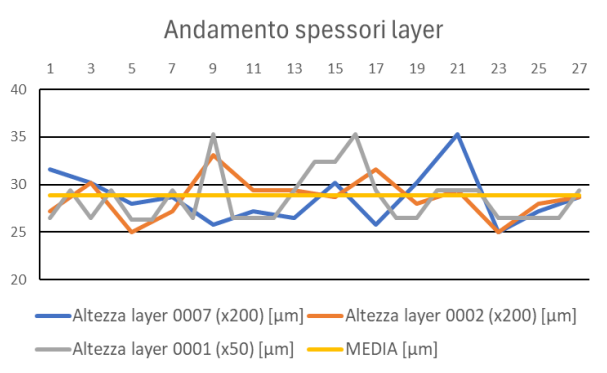
By comparing the green specimen data with the average of the sintered ones, the percentage shrinkage of the specimen can be estimated. Before sintering, the resin is removed from the specimen through the *debinding* process, after which the temperature can be raised to obtain a ceramic part. Percentage shrinkage is defined as the ratio between the difference of the two measurements and the initial measurement.

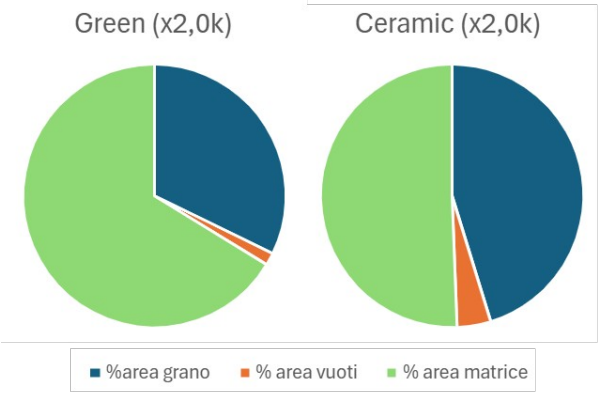
Flexural bars	*Length*	*Width*	*Height*	*Area*	*Volume*
(CAD-Green)/CAD	1.5%	-8.2%	12%	1.0%	6.2%
(Green-Ceramic)/Green	17.1%	15.5%	10.2%	28.4%	37.1%
(CAD-Ceramic)/CAD	18.3%	8.6%	21%	29.1%	41.0%

[3] Specimen preparation for scanning electron microscopy: for the specimen to be observed with the SEM, it must be electrically conductive; therefore, the specimen surface is coated with a gold layer about 40 nm thick by sputtering in an argon plasma inside a vacuum chamber.





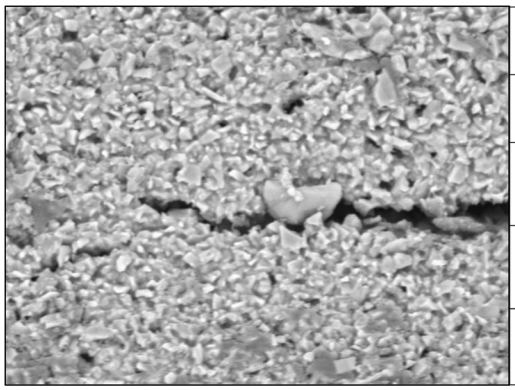




The layer height was measured from three SEM images at different magnifications using a purpose-written MATLAB® script. A statistical analysis, which was also used to properly calibrate the script, yielded a layer height of 28.836 ± 0.332 µm, against a nominal printing layer height of 30 µm. [4]

By analysing images at different magnifications, a statistical estimate of grain and void size was obtained. The grain axis increases from 0.9335 ± 0.0297 µm to 0.9767 ± 0.079 µm, while the void axis increases from 0.5034 ± 0.0172 µm to 0.7615 ± 0.0269 µm: sintering led to material transport and fusion between neighbouring particles, enlarging both grains and voids. By properly calibrating the script, the percentages of grains, matrix and voids were estimated. The matrix fraction decreases from the green to the ceramic specimen, although part of it remains in the ceramic one. A higher void content is also observed in the ceramic specimen compared with the green one.

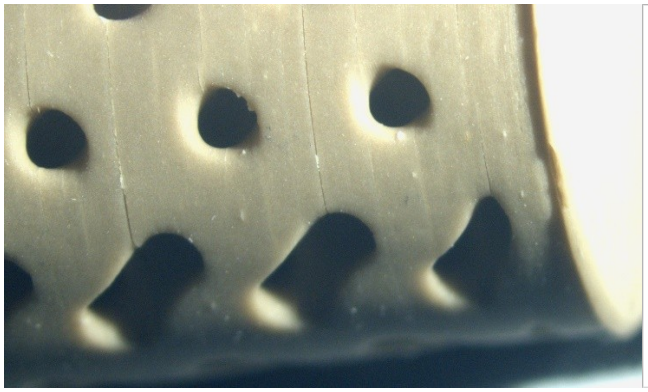
SEM analysis reveals cracks in both the green and the sintered specimens. In the green specimen, larger particles drive crack propagation, helped by the fact that the particles are not fused together. Cracks in the ceramic specimen are caused by large regolith grains that did not fuse with the others during sintering.

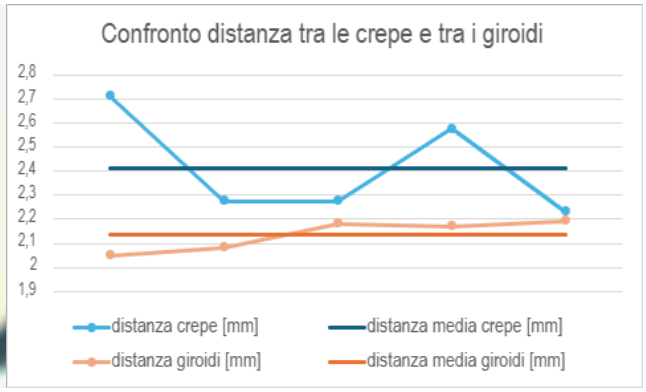


Area [µm²]	70,877
Perimetro [µm]	37,289
Asse maggiore [µm]	13,7530
Asse minore [µm]	7,0284
Eccentricità [-]	0,85991

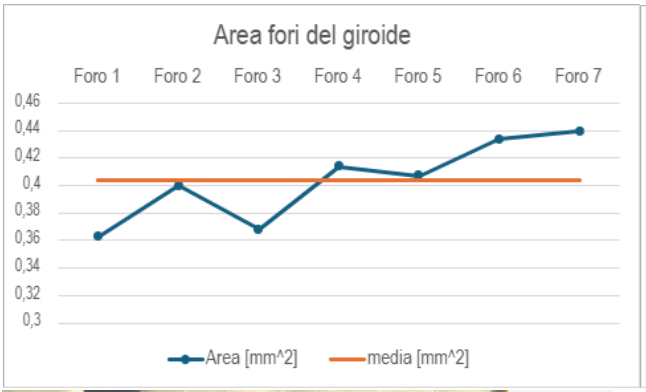
[4] The uncertainties reported in this section are computed as the ratio between the sample standard deviation and the square root of the number of observed samples, since they derive from a statistical analysis of the collected data.

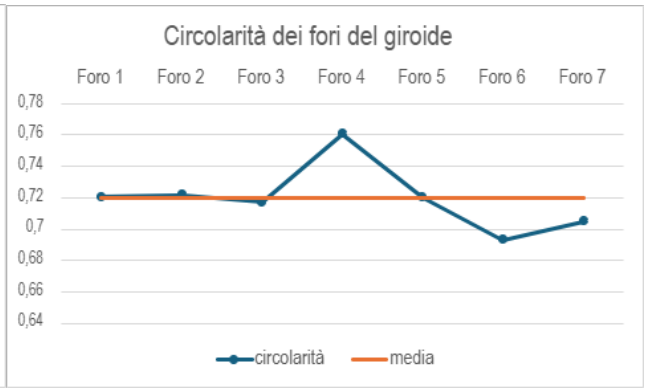
These grains drive crack propagation, triggered by stress concentrations during the sintering process. An example of a large grain is shown in the figure. The grain has a major axis of 13.753 µm and is considered large compared with the maximum measured regolith powder size of about 8.710 µm. The resulting crack is longer than 1882 µm (a limit set by the image size itself). The average thickness of the observed cracks is 13.098 µm (no uncertainty is given, since the collected measurements did not allow a meaningful statistical analysis).
Images of the sintered cylinders show cracks at regular intervals, comparable to the spacing between the holes generated by the gyroid. This is assumed to be caused by the low material fraction in the corresponding layer.

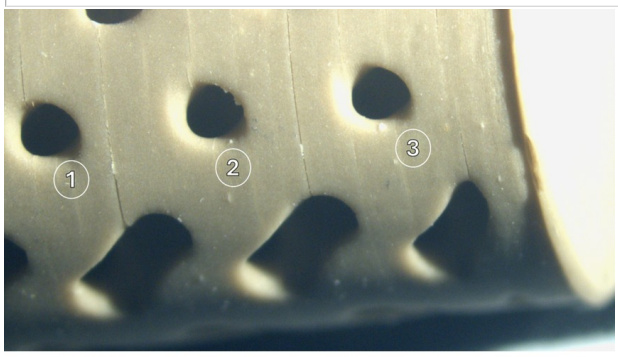




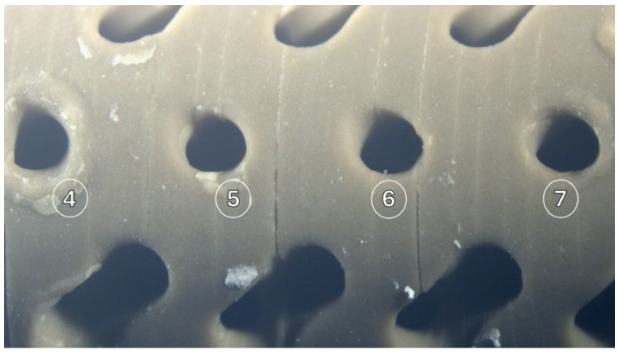
From the optical microscope images, the area and circularity of the gyroid holes of the two sintered cylinders can be measured.







5A_sin_5_1.1x

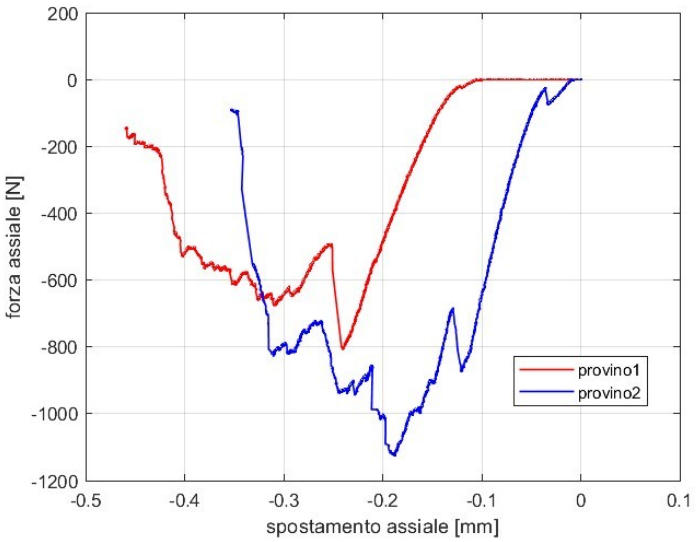


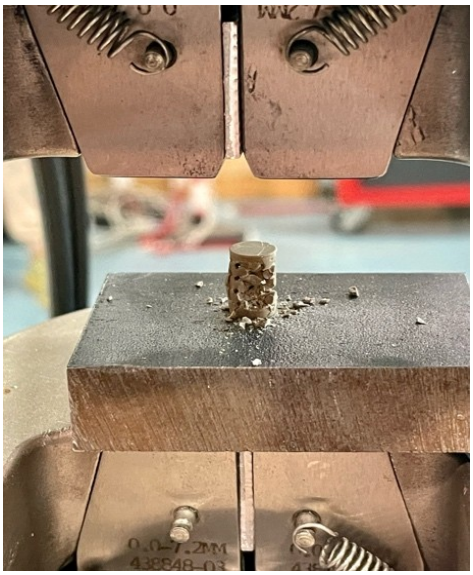
5B_ceram_1_1.1x

3.2. Mechanical Properties

To evaluate the mechanical properties of the manufactured components, compression and flexural tests were carried out in the laboratory. Two gyroid-lattice cylinders were tested in

compression up to failure using the *MTS 858 Mini Bionix II®* system, recording the applied axial force and the vertical displacement over time (plotted below).

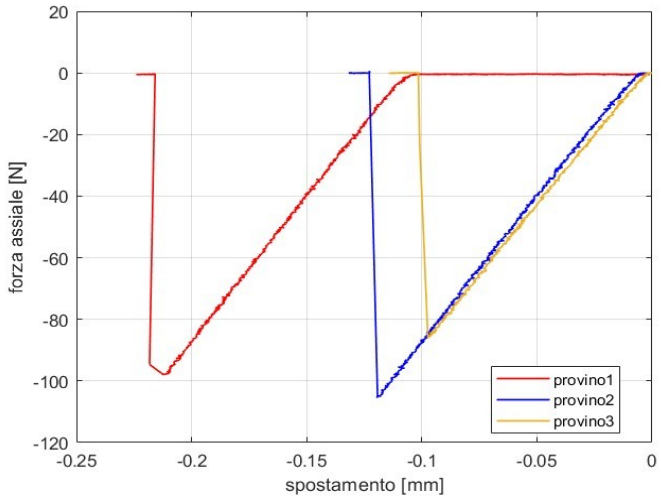




The data were then processed according to the ASTM C1424−25 standard to compute the compressive strengths of the specimens, equal to 109.8726 MPa and 153.0380 MPa, respectively. Since the tested cylinders have a 60% “gyroid infill”, the load-bearing area considered is the minimum one, equal to 60% of the nominal area. In addition, the stiffness values in the linear regions were estimated: 7.2938 kN/mm and 10.967 kN/mm. The stress-strain curve was not derived, as no further data were available.

Three bar specimens were tested in flexure up to failure using the “Three-Point Fixture” configuration described in the ASTM C1161−18 standard.

The data were used to compute the flexural strength with the standard formula given by the standard for the selected configuration. The corresponding curves and the flexural strength values of the three tested specimens are reported below.



Flexural strength
Specimen 1
S1 = 134.4267 MPa
Specimen 2
S2 = 147.9392 MPa
Specimen 3
S3 = 114.9294 MPa

In both cases, a statistical analysis of the mechanical properties was not possible due to the limited number of available specimens.

4. Discussion and Conclusions

The quality analysis led to a detailed investigation of the cracks that developed in the manufactured parts.
The defects were studied systematically through an Ishikawa (fishbone) diagram, from which the human factor was excluded. Crack formation is related to:

- Machine error
  Incomplete detachment from the build plate and possible mechanical stresses in the part may result from excessive *lift* and *retract* speeds.
- Method error
  Incomplete curing may be caused by insufficient exposure time and/or too short waiting times.
- Material error
  The small particle size increases the likelihood of agglomeration-related printing defects.

The errors described above lead to less accurate components, highlighting the limits of the adopted manufacturing process. Moreover, the manual definition of the image-processing script parameters, although supported by a statistical analysis, is a possible limitation in the acquisition of measurement data.

All these issues, together with random and non-removable errors, make the process not perfectly repeatable, as also shown by the inconsistency between the flexural failure points of the bars and by the different compressive behaviour of the cylinders.
A mechanical test campaign with a statistically significant number of specimens could provide more robust indications.

Future work could investigate how print orientation, layer thickness and gyroid size affect the mechanical properties and quality of the printed parts.

5. References

1. Characterization of planetary regolith simulants for the research and development of space resource technologies, Jared M. Long-Fox, Daniel T. Britt
   (Original Research article, Front. Space Technol., 27 September 2023, Sec. Space Exploration)

2. The influence of particle size distribution on the curing behavior of ceramic-filled resins for vat photopolymerization, Author: Wadih Yared, Rainer Gadow, Version of Record 7 June 2023

3. DATASHEET: *Elegoo Mars 4 DLP®*

4. Effect of particle size on rheology, curing kinetics, and corresponding mechanical and thermal properties of aluminum nitride (AlN) ceramic by digital light processing (DLP)-based vat photopolymerization Authors: Lifu Lin, Haidong Wu, Yehua Li, Junye Wang, Shanghua Wu, Journal of the European Ceramic Society, Volume 44, Issue 1, January 2024, Pages 184-192

5. C1161 – 18 (Reapproved 2023): Standard Test Method for Flexural Strength of Advanced Ceramics at Ambient Temperature

6. C1424 – 25: Standard Test Method for Monotonic Compressive Strength of Advanced Ceramics at Ambient Temperature

7. Image Processing Onramp® (MathWorks®)

8. Image Processing with MATLAB® (MathWorks®)